

1. Administrative & Study Identification

Full Study Title: Real World Impact and Effectiveness of ABRYSVO® Vaccination During Pregnancy Against RSV Illness in Infants **Short Title/Acronym:** BERNI (Impact and Effectiveness of ABRYSVO® Vaccination During Pregnancy) **Protocol Number:** C3671068 **NCT Number:** NCT06647654 **Sponsor Name:** Pfizer **Investigational Product:** Biological: ABRYSVO (bivalent RSV prefusion F protein-based vaccine / RSVpreF) **Phase of Development:** Observational / N/A (Retrospective Study) **Medical Monitor:** Pfizer (Responsible Party)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To estimate the vaccine effectiveness (VE) of maternal ABRYSVO vaccination during pregnancy against RSV-associated lower respiratory tract disease (LRTD) hospitalization in infants. **Secondary Objectives:** To evaluate VE estimates stratified by several infant/maternal characteristics; to describe infant characteristics, severity/clinical features, and healthcare resource use during the index hospitalization; and to evaluate the population-level impact of ABRYSVO introduction on RSV-associated and all-cause outcomes among infants (Ecologic Before-and-After Design).

2.2 Background & Rationale This non-interventional retrospective study is being conducted to evaluate the real-world impact and effectiveness of the bivalent RSV prefusion F protein-based vaccine (ABRYSVO) administered during pregnancy. The study aims to understand how well the recently implemented maternal ABRYSVO program protects infants from severe RSV illness and hospitalization during their most vulnerable early months of life.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Test-Negative Design and Ecologic Before-and-After Design) **Control Type:** Test-negative controls (for TND) and historical pre-ABRYSVO RSV seasons (for Ecologic Design) **Blinding:** None (Open-label / Retrospective analysis) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: Retrospective inclusion of eligible hospitalized infants meeting LRTD criteria at participating sites. **Number of Sites:** Participating hospital sites in Argentina and Uruguay. **Actual Study Start:** 26-Aug-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Infants ≤ 9 months (≤ 270 days) of age on the index date (for the Test-Negative Design) or infants and children ≤ 24 months of age (for the Ecologic Before-and-After Design). 2. Infants born at 32 weeks of gestational age or greater. 3. Admitted to one of the participating hospital sites with symptoms of respiratory infection, meeting the definition of LRTD, and having a respiratory specimen collected with an RSV test result through standard-of-care testing.

4.2 Exclusion Criteria 1. Received any licensed or investigational RSV preventive product (e.g., Palivizumab, Nirsevimab, active RSV vaccine) since birth, or received ≥ 1 blood transfusion/blood products containing antibody since birth. 2. Born to an individual who received any other licensed or investigational RSV vaccine during pregnancy, or for whom ABRYSVO vaccination status cannot be confirmed in available data sources. 3. Infants with LRTD that require hospitalization for reasons other than clinical criteria (e.g., for social reasons).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: ABRYSVO vaccine administered as a single maternal dose according to the standard of care during pregnancy (prior to the study's infant index date). **Route of Administration:** Intramuscular (administered to the pregnant individual). **Duration of Treatment:** N/A (Retrospective study assessing the protective impact of maternal vaccination on infants up to 9–24 months of age).

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care medical treatments for pediatric LRTD during hospitalization. **Prohibited:** Use of Palivizumab, Nirsevimab, other active RSV vaccines, or blood products containing antibodies post-birth (results in exclusion from the analysis cohort).

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Historical Data** | Pre-ABRYSVO Seasons |
Aggregated baseline data collection for ecologic comparison (pre-2024 RSV seasons). | | **Index Hospitalization** | RSV Seasons (Apr - Sep) | Retrospective extraction of symptoms, severity/clinical features, RSV test results, and healthcare resource use.
| | **Post-ABRYSVO** | 2024-2026 Seasons | Hospitalization data

extraction, verification of maternal vaccination status, and final VE analysis. |

(Note: As this is a retrospective observational study, there are no prospective patient visits.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Vaccine Effectiveness (VE) of maternal ABRYSV0 vaccination against RSV-associated LRTD hospitalization. **Measurement Method:** Calculated via Adjusted Odds Ratio (aOR), comparing the odds of maternal ABRYSV0 vaccination during pregnancy between test-positive cases and test-negative controls. VE is estimated as $(1 - aOR) \times 100\%$.

7.2 Secondary & Exploratory Endpoints * VE estimates stratified by various demographic and clinical characteristics. * Rates of RSV-associated and all-cause outcomes among infants ≤ 6 months (compared with older age groups) over multiple RSV seasons overall and by epidemiological week, calendar year, calendar month, and hospital site (Ecologic Before-and-After evaluation).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospective chart review of severity/clinical features and the use of healthcare resources during index hospitalizations for LRTD. **Serious Adverse Events (SAEs):** N/A (Standard reporting timelines do not apply to this retrospective observational data collection). **Data Monitoring Committee (DMC):** N/A for non-interventional retrospective design.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Test-negative hospitalized infants ≤ 9 months (TND study cohort) and infants/children ≤ 24 months (Ecologic study cohort). **Sample Size Justification:** Utilizes all eligible hospitalized infants mapped during the defined periods; derived from available aggregated real-world data without explicit prospective powering. **Handling of Missing Data:** Exclusion of cases where maternal ABRYSV0 vaccination status cannot be confirmed from available data sources.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP), applicable real-world evidence guidelines, and the Declaration of Helsinki. **Monitoring Plan:** Source data verification and retrospective database/

Electronic Medical Record (EMR) query validation. **Audit Trail:** Secure data abstraction from medical records into validated Electronic Data Capture (EDC) systems.

11. Study Identifiers & Governance

Primary ID: C3671068 **Registry Number:** NCT06647654 **Sponsor/Lead Organization:** Pfizer **Study Title:** BERNI (Impact and Effectiveness of ABRYSV0® Vaccination During Pregnancy) **Official Regulatory Title:** Real World Impact and Effectiveness of ABRYSV0® Vaccination During Pregnancy Against RSV Illness in Infants

12. Clinical Framework (Study Specific)

Primary Condition: Respiratory Syncytial Virus (RSV) / Respiratory Tract Diseases **Diagnosis Threshold:** Hospital admission with symptoms of respiratory infection and positive RSV testing (LRTD criteria) **Medical Methodology:** Test-negative design (TND) and Ecologic before-and-after retrospective analysis **Optimization Target:** Infant protection via maternal ABRYSV0 vaccination

13. Study Procedures & Methodology

Management Pathway: Retrospective assessment of standard clinical pathway for infant LRTD **Education Protocol (HCP):** N/A (Observational / Retrospective) **Education Protocol (Patient):** N/A (Observational / Retrospective) **Quality Audit Mechanism:** EMR chart review and data abstraction validation

14. Observation & Follow-Up Schedule

Total Duration: Retrospective tracking of infant hospitalizations up to 9 months (TND) or 24 months (Ecologic) of age **Onsite Visit Cadence:** N/A (Retrospective extraction of hospitalization records) **Remote Monitoring Frequency:** N/A (Database-driven approach) **Checklist Review Interval:** Data extracted across RSV seasons (April to September in 2024, 2025, and 2026)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				